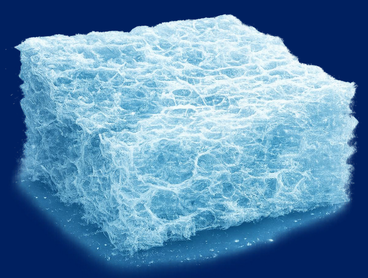
SUSTAINABLE THERMAL PROTECTION MATERIALS



Low-Cost Bacterial Cellulose Aerogels for Lightweight Thermal Protection in Spacecraft Heat Shields

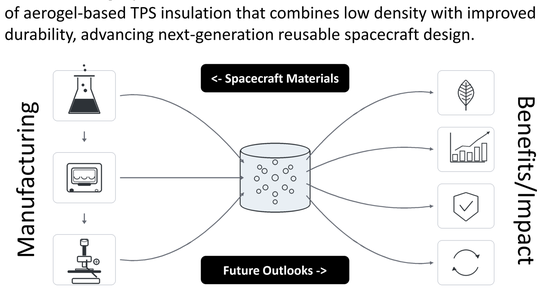
Amogh Vinaykumar, Flower Mound High School

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SUMMARY

With rapid space exploration and the need for reusable spacecraft, lightweight and robust thermal protection systems (TPS) are critical. Existing intermediate-layer insulators often suffer from thermal degradation, mechanical fragility, and added mass under repeated reentry conditions. This project develops a bio-derived bacterial cellulose aerogel enhanced with citric acid, montmorillonite clay, and polyvinyl alcohol. Samples are fabricated through a process that involves hydrogel casting, thermal curing, solvent exchange, and lyophilization to preserve their structural integrity. This work introduces a scalable and sustainable class of aerogel-based TPS insulation that combines low density with improved durability, advancing next-generation reusable spacecraft design.



INTRODUCTION

Spacecraft heat shields are the critical barrier that protect vehicles from enduring the harsh environmental pressures that are faced during reentry.

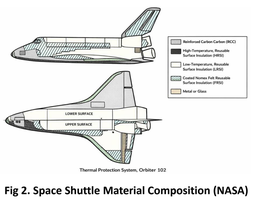
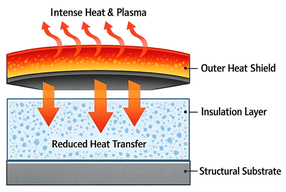


Fig 2. Space Shuttle Material Composition (NASA)

Fig 3. Thermal Protection System Layered Stack



Modern thermal protection systems are built as a layered stack, where each layer serves a distinct function.

- Outer layer (extreme surface heating) -> Intermediate layer (slows heat conduction and stabilizes) -> Substrate (keeps everything attached)

Ceramic Fiber Blankets: good thermal stability, but relatively heavy and prone to degradation under repeated thermal cycling

Silica Aerogels: low density & exceptional insulation, but inherently brittle and mechanically fragile

Polymeric Foam: lightweight, with good initial insulation, but limited temperature tolerance and long-term thermal durability

Knowledge Gap: There is a need for a lightweight, thermally efficient insulation material that maintains mechanical robustness and durability under extreme spaceflight conditions.

What are aerogels? Aerogels are low-density, highly porous solids formed by removing a gel's liquid phase without collapsing its structure, resulting in exceptional thermal insulation and minimal mass.

Why bacteria cellulose? Through a renewable nanofiber network, bacterial cellulose enables reduced brittleness and stress while supporting scalable and sustainable material production

Key Equation: Fourier's Law (1D heat conduction)

$$q'' = k\,\Delta T / L$$

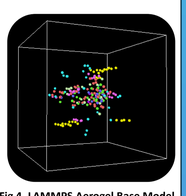


Fig 4. LAMMPS Aerogel Base Model

OBJECTIVES AND PURPOSE

Engineering Problem: Current spacecraft thermal protection systems rely on intermediate and outer-layer materials that provide **excellent thermal insulation** but often involve **high** material costs, **energy-intensive** manufacturing processes, **increased mass**, and **limited long-term** sustainability and scalability.

Research Question: Can a bacterial cellulose (BC) aerogel, prepared with multiple enhancers, surpass the **performance** of conventional insulation material while offering a sustainable alternative to this costly application?

Claim/Goal: This research will develop a **BC aerogel** that demonstrates the ideal blend of properties for **aerospace applications**, thereby improving environmental sustainability compared to petrochemical-based alternatives. This will establish a **pathway** toward a new class of lightweight, mechanically robust insulators with potential **impact** across insulation applications **worldwide**.

Engineering Design Criteria

Thermal Efficiency: Achieves ultra-low thermal conductivity & stability under extreme reentry heat loads

Lightweight Structural Integrity: Maintains mechanical strength and durability while minimizing mass

Sustainable & Scalable Design: Derived from renewable materials using energy-efficient, scalable processes

Purpose & Impact – Why this project and what are the outlooks?

- ***Novelty:*** Addresses research gap in sustainable, robust aerospace thermal insulation materials (especially for overall aerogel applications)
- ***Reusability:*** Enables frequent spacecraft missions through durable, lightweight protection
- ***Efficiency:*** Reduces manufacturing environmental impact & cost while meeting extreme reentry thermal-mechanical demands

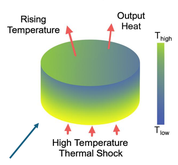
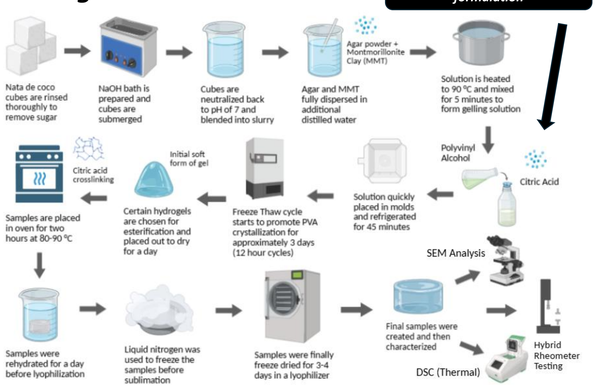


Fig 5. Aerogel Heat Dissipation Representation

ENGINEERING METHODOLOGY

Aerogel Creation Process

Optimization varies with formulation



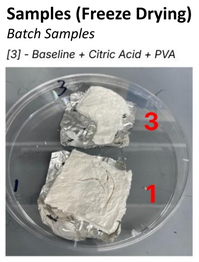
Experimentation Summary: Simulation & Modeling -> Precursor Preparation -> Sol-Gel Processing & Enhancer Dispersion -> Homogenization -> Freeze Thaw Cycle -> Esterification -> Freeze Drying -> Characterization

- **Bacterial cellulose** extracted from nata de coco, cleaned with NaOH, and homogenized into a uniform slurry before sol-gel casting
- **Freeze-thaw cycling**, fridge (~2 °C) to freezer (~-20 °C), induces PVA crystallization, while citric acid oven esterification (~100 °C) consolidates the polymer network
- **Final hydrogels** are freeze-dried via first liquid nitrogen freezing and then lyophilized by a Millrock benchtop dryer (Rice University) for ~3 days; additionally, a few samples underwent a cheap, alternative ambient pressure drying process conducted at home to compare results

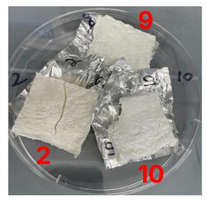
Samples (Freeze Drying)

Batch Samples

[3] - Baseline + Citric Acid + PVA



[1] - Baseline (agar + nata de coco)



[2] - Baseline (agar + nata de coco)
[9] - Baseline + Citric Acid + PVA + Low clay (esterified sample)
[10] - Baseline + Citric Acid + PVA + High clay (esterified sample)

[4] - Baseline + Citric Acid + PVA + Low Clay (damaged)

[7] - Baseline + Citric Acid + PVA + extra citric acid soak (esterified sample)
[8] - Baseline + Citric Acid + PVA (esterified sample)

Figure 9. Front and Side Profile of Samples

Fig 6. Sol-Gel Solution



Citric Acid Crosslinking & Esterification

- Reagent-grade citric acid forms covalent ester bonds between carboxyl (-COOH) and hydroxyl (-OH) groups on cellulose chains, creating a 3D polymer network
- Thermal activation causes a condensation reaction to occur, where water is released and ester bonds (-COO-) form, enhancing structural stability

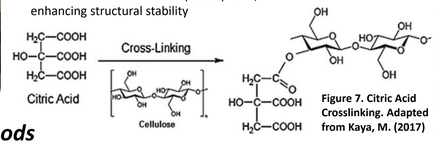
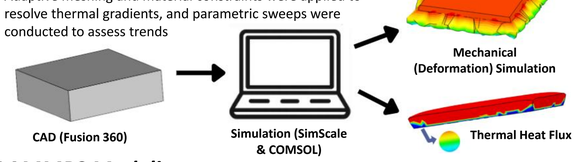


Figure 7. Citric Acid Crosslinking. Adapted from Kaya, M. (2017)

Simulation Methods

- 50 × 50 × 20 mm CAD models in Fusion 360 were prepared to define the aerogel domain and mimic experimental sample geometry
- Adaptive meshing and material constraints were applied to resolve thermal gradients, and parametric sweeps were conducted to assess trends



LAMMPS Modeling

- Citric acid provides a strong network and connection between molecules
- Ester crosslinking complements PVA reinforcement
- Montmorillonite clay groups together to distribute strength and support nata de coco

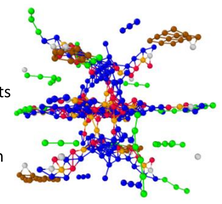


Figure 8. BC Aerogel LAMMPS Model

Bacterial Cellulose
Citric Acid Crosslinker
Montmorillonite Clay
PVA Polymer
Ester Crosslink
Pore Structure

Preparation and Imaging

- Surfaces and cross sections of samples were measured and prepared before analysis
- Samples were then coated with 10 nm gold using a sputter coater
- SEM images were then collected at multiple depths and magnifications revealed spatial variations in pore structure, enabling a model of the aerogel's internal architecture

SEM Micrographs & Structure

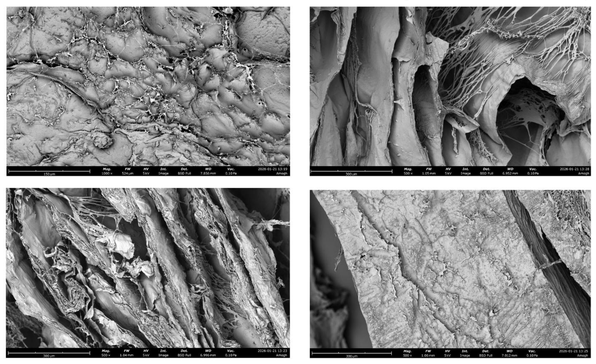


Figure 10. Sputter Coater Process

RESULTS & KEY FINDINGS

Figure 11. DSC Thermogram of Bacterial Cellulose Aerogel Samples

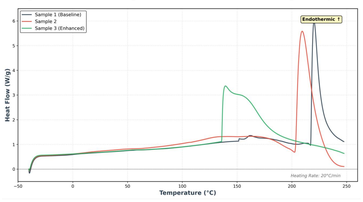


Figure 12. Sample Thermal Profiles Over Time

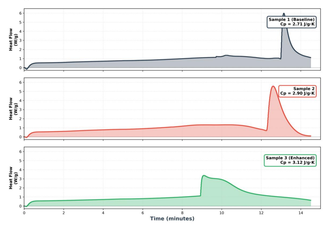


Figure 13. Thermal Performance Heat Map

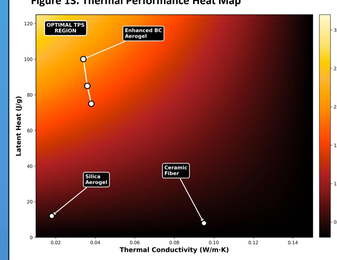
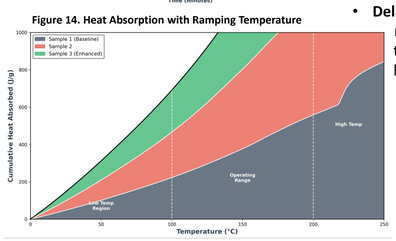


Figure 14. Heat Absorption with Ramping Temperature



Thermal Characterization

- **Preparation:** Sample was hermetically sealed in a Tzero aluminum pan and analyzed using *Differential Scanning Calorimetry (DSC).*
- **High specific heat capacity:** Cp ranged from 2.71 → 3.12 J/g·K, representing a 3.4–3.7× improvement over silica aerogel and ceramic fiber insulation.
- **Strong thermal buffering:** Stable heat flow (0.6–1.0 W/g) maintained from 25–250°C cycle, matching intermediate TPS operating temperatures (~140–250°C).
- **Energy Absorption:** Citric acid crosslinking unlocks a broad endothermic region (130–180°C), absorbing ~80–100 J/g prior to any cellulose degradation.
- **Delayed heat penetration:** Estimated thermal diffusivity α ≈ 1.6 × 10^{-7} m^2/s from data collection + relevant literature, indicates slow temperature rise under transient heating.

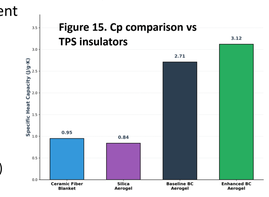
$$C_p = \frac{\dot{q}_{\text{sample}}}{m \cdot \beta}$$

β = heating rate (K·s^{-1})

$$\alpha = \frac{k}{\rho C_p}$$

α = thermal diffusivity (m^2/s)
k = thermal conductivity (W/m·K)
ρ = density (kg/m^3)
Cp = specific heat capacity (J/kg·K)

Figure 15. Cp comparison vs TPS insulators



Von Mises Stress (N/m^2) Test

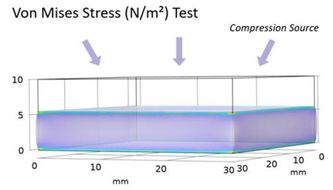
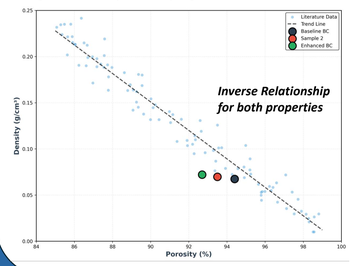
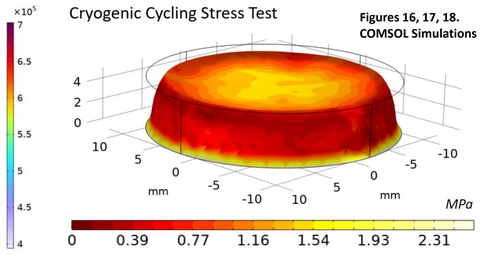


Figure 19. Porosity vs Density Regression

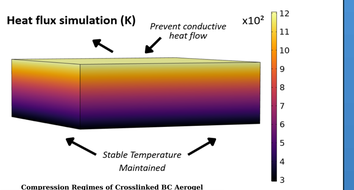


Cryogenic Cycling Stress Test



Figures 16, 17, 18. COMSOL Simulations

Heat flux simulation (K)



Simulation and General Properties

- **Multiphysics simulations (COMSOL, SimScale)** modeled many realistic conditions, revealing strong radiative shielding, cryogenic stability, and reduced backside conductive heat flow above 400 °C.
- **Hybrid Rheometer testing** confirms the crosslinked BC aerogel achieves a compressive modulus of ~179.2 kPa with a peak stress of 59.6 kPa, representing a 7.9× and 4.5× improvement over uncrosslinked baselines
- **SEM/ImageJ analysis** revealed high porosity (~94.5% baseline; ~92.7% enhanced) with nanoscale pore sizes (0.02–0.04 μm), consistent with effective thermal insulation in many aerogel materials.
- **Density and porosity calculations** $\rho = m/V$, $\phi = 1 - \rho_{\text{bulk}}/\rho_{\text{solid}}$ yielded densities of ~0.064 g/cm³ (baseline) and ~0.078 g/cm³ (enhanced), balancing integrity and weight.

Figure 23. Compression Regimes From Hybrid Rheometer Analysis

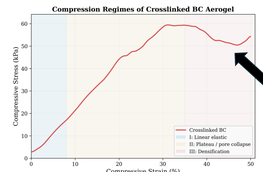
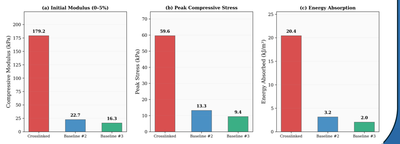


Figure 24. Batch Variation Mechanical Testing



STATISTICAL ANALYSIS

Statistical analysis combined experimental replicates (n ≥ 3 per formulation) with derived estimates using reproducible processing, error quantification, and sensitivity tools. Both parametric and nonparametric tests, along with uncertainty propagation, were employed to evaluate trends rigorously.

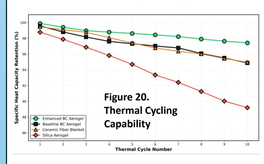
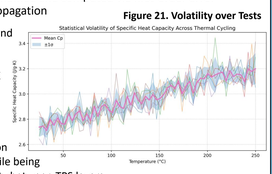


Figure 20. Thermal Cycling Capability

- **Raw DSC and density data** were normalized, baseline-corrected, and integrated consistently (results were reported as mean ± SD).
- **Places of uncertainty** were through instrument resolution, sample geometry, and variability, so bootstrap resampling (10,000 draws) was used for robust confidence intervals.
- **For the derived properties** (thermal conductivity, CTE, α), both experimental data and literature were combined and computed via first-order error propagation

Figure 21. Volatility over Tests



- **Normality** was checked through the Shapiro–Wilk test and then ANOVA + Tukey HSD compared groups, with Cohen's d reported to show practical significance
- **Thermal conductivity** was estimated to **k=0.035 W/m·K** (low heat transfer due to Knudsen effect -> reduced gas conduction due to tiny pores)
- **The shrinkage** for the enhanced aerogel was **<10%**, better than the typical 15–25% in silica aerogels.
- **Through simulation**, the coefficient of thermal expansion (CTE) ≈ **1.2 × 10^{-6} K^{-1}**, which was similar to ceramics while being much lower than metals, showcased strong compatibility between TPS layers.

Table 1. Application Comparison

Property	Baseline BC Aerogel	Enhanced BC Aerogel	Silica Aerogel	Ceramic Fiber Blanket
Manufacturing Cost (per sample)	~$6	~$10	~$50-100	~$75
Structural Integrity	Slightly Brittle	Semi-Ductile	Very Brittle	Flexible
Environmental Impact	Biodegradable Renewable	Biodegradable Renewable	Non-biodegradable Energy-intensive	Non-biodegradable Carcinogenic fibers
Conformability	Rigid	Semi-Flexible	Rigid	Highly Flexible

DISCUSSION & CONCLUSIONS

This study addresses four critical barriers in intermediate TPS design: sustainability, scalability, durability/thermal performance, and accessibility, by establishing a bio-derived bacterial-cellulose aerogel architecture. By introducing a lightweight, sustainable alternative, this work opens a new design space for next-generation spacecraft reentry systems and reframes how thermal protection materials can be engineered.

- **Sustainability:** Renewable bacterial cellulose and citric-acid chemistry eliminate petroleum precursors and supercritical drying, reducing energy consumption and avoiding carcinogenic, unsustainable processing used in ceramic fibers.
- **Scalability:** Simple, repeatable processing (homogenization, solvent exchange, freeze-drying) with tunable parameters enables solvent recycling and realistic scale-up to TPS criteria.
- **Durability & thermal performance:** High heat capacity (≈3.1 $J \cdot g^{-1} \cdot K^{-1}$), durable cycling performance, and low thermal diffusivity (~1.6×10^{-7} $m^2 \cdot s^{-1}$) provide effective transient heat buffering; samples show little to no deformation for temperatures >200 °C.
- **Accessibility:** Lower-cost equipment/procedures and safer chemistry than traditional aerogels broaden adoption, with formulations tunable for mission-specific thermal needs.

Revisiting Criteria and Claim:

- **Tradeoffs in formulation:** Submerging (P3) in an extra citric acid round produced the best thermal buffering, strength, and low shrinkage, while too much clay (P4) reduced porosity and caused brittleness.
- **Balanced performance:** BC aerogel stores 3–4× more heat than silica aerogel but conducts less than most ceramic fibers, combining the benefits of both.
- **Affordable fabrication:** Simple methods like dry-ice freezing and ambient-pressure drying can produce near-lyophilized samples without expensive equipment.
- **Wide relevance:** By collaborating with many researchers and grounding the work in extensive literature, this material serves as a model for not only spacecraft insulation but also as a pathway for advanced aerogel applications

Figure 22. Aerogel Applications



Overall, the results highlight BC aerogels as a promising class of materials whose tunable structure enables continued innovation, bringing advanced space technologies one step closer to Mars.

FUTURE WORK

The results demonstrate a novel and impactful exploration of **bacterial cellulose aerogels for thermal protection**, but this work represents only the beginning, which future studies will explore

- **Environmental validation**: Test samples under repeated cryogenic cycling, vacuum outgassing, and radiative heating to assess thermal shock resistance and heat-transfer stability, mimicking space environment conditions.
- **Shape & integration**: Develop conformal aerogel geometries for curved heat shields/evaluate thermo-mechanical behavior under compressive loading.
- **Advanced simulation**: Conduct higher-fidelity Multiphysics modeling of reentry-like radiative scattering, heat flux, infrared emissivity, and property relationships based on porosity and filler distribution.

Long-term (equipment/budget limited)

- Explore hybrid aerogel architectures capable of withstanding outer heat-shield conditions, including **UHTC-based reinforcements** (e.g., hafnium carbide)
- Expand on strengthening BC aerogel design through **carbonization** and **nanoparticle incorporation** (e.g., zinc borate, ammonium phosphate).
- Broaden thermal insulation **applications** in **aerospace, construction, and industrial systems**, optimizing phonon scattering, radiative suppression, and energy management.

KEY REFERENCES

[1] Jin, R., Zhou, Z., Liu, J., Shi, B., Zhou, N., Wang, X., Jia, X., Guo, D., & Xu, B. (2023). Aerogels for Thermal Protection and Their Application in Aerospace. *Gels*, *9*(8), 606. https://doi.org/10.3390/gels9080606

[2] Phan, H. T., Nguyen, K. D., Nguyen, H. H. M., Dao, N. T., Le, P. T. K., & Le, H. V. (2023). Nata de coco as an abundant bacterial cellulose resource to prepare aerogels for the removal of organic dyes in water. *Bioresource Technology Reports*, *24*, 101613. https://doi.org/10.1016/j.biteb.2023.101613

[3] Aegerter, M. A., Leventis, N., & Koebel, M. M. (Eds.). (2011). *Aerogels Handbook*. Springer New York. https://doi.org/10.1007/978-1-4419-7589-8

[4] Sozcu, S., Frajova, J., Wiener, J., Venkataraman, M., Tomkova, B., & Militky, J. (2024). Effect of Drying Methods on the Thermal and Mechanical Behavior of Bacterial Cellulose Aerogel. *Gels*, *10*(7), 474. https://doi.org/10.3390/gels10070474

[5] Revin, V. V., Nazarova, N. B., Tsareva, E. E., Liyaskina, E. V., Revin, V. D., & Pestov, N. A. (2020). Production of Bacterial Cellulose Aerogels With Improved Physico- Mechanical Properties and Antibacterial Effect. *Frontiers in Bioengineering and Biotechnology*, *8*, 603407. https://doi.org/10.3389/fbioe.2020.603407

[6] Seiferth, Rolf. *Ablative Heat Shield Design for Space Shuttle*.

Paper bibliography contains full list.